

Unit 9, Lesson 5: Rotations on the Coordinate Plane

Benchmark

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

Learning Targets

- ☐ I can perform a rotation on the coordinate plane.
- ☐ I can describe relationships between figures after a rotation.

9.5.1 Warm-Up: Geometry in Art

Tania was a Jewish American artist who was born in Poland and based in New York City. She was an abstract painter, sculptor, and collage artist. She was famous for painting wall art throughout the city.

One of her most well-known pieces of wall art was a 13-story geometric wall painting she created in 1970 that still stands today. It is shown in the picture to the right.

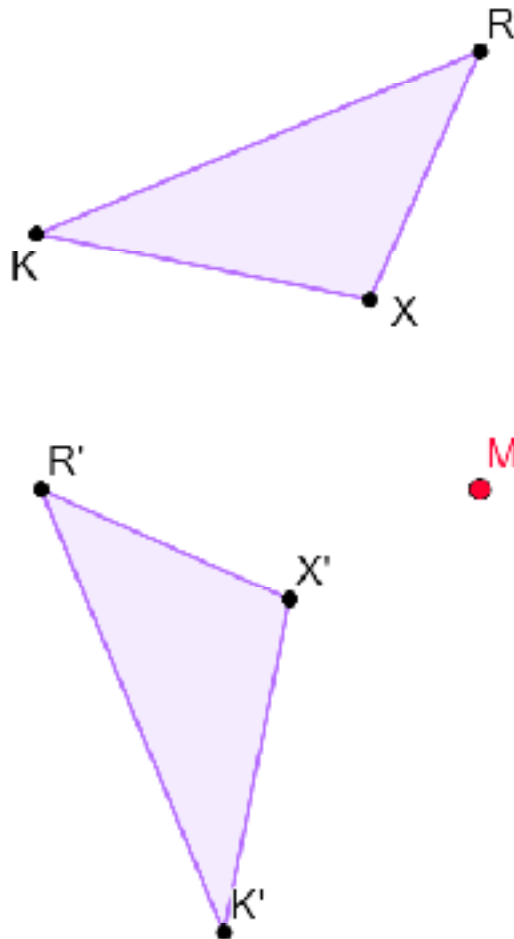


1. Consider the yellow and green triangles shown in the artwork.
 - a. Explain whether you think a single transformation could map the yellow triangle to the green triangle.
 - b. Which two transformations do you think could map the yellow triangle to the green triangle if one was performed followed by the other?

9.5.2 Exploration: Determining the Degree of Rotation

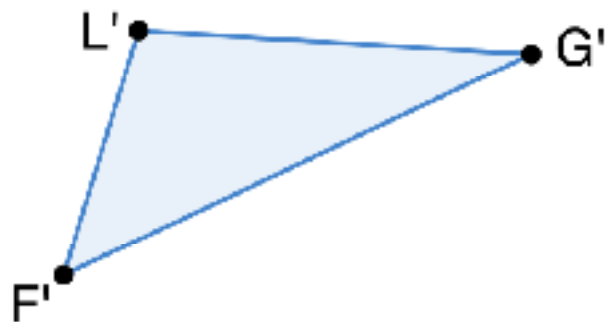
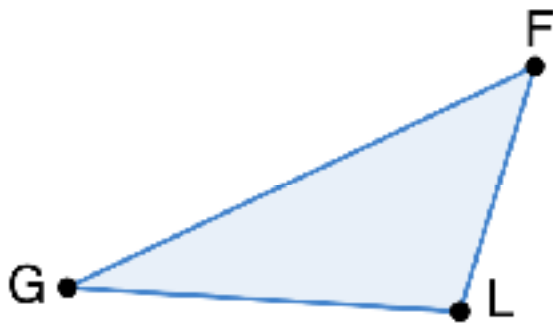
Work with your partner to complete the following.

1. The diagram shows $\triangle RXX$ and $\triangle R'X'X'$ after a rotation about point M .



- a. Use a straightedge to draw $\angle XMX'$, $\angle RMR'$, and $\angle KMK'$.
- b. Discuss with your partner how the degree of rotation can be determined using the angles that connect the vertices and the center of rotation.

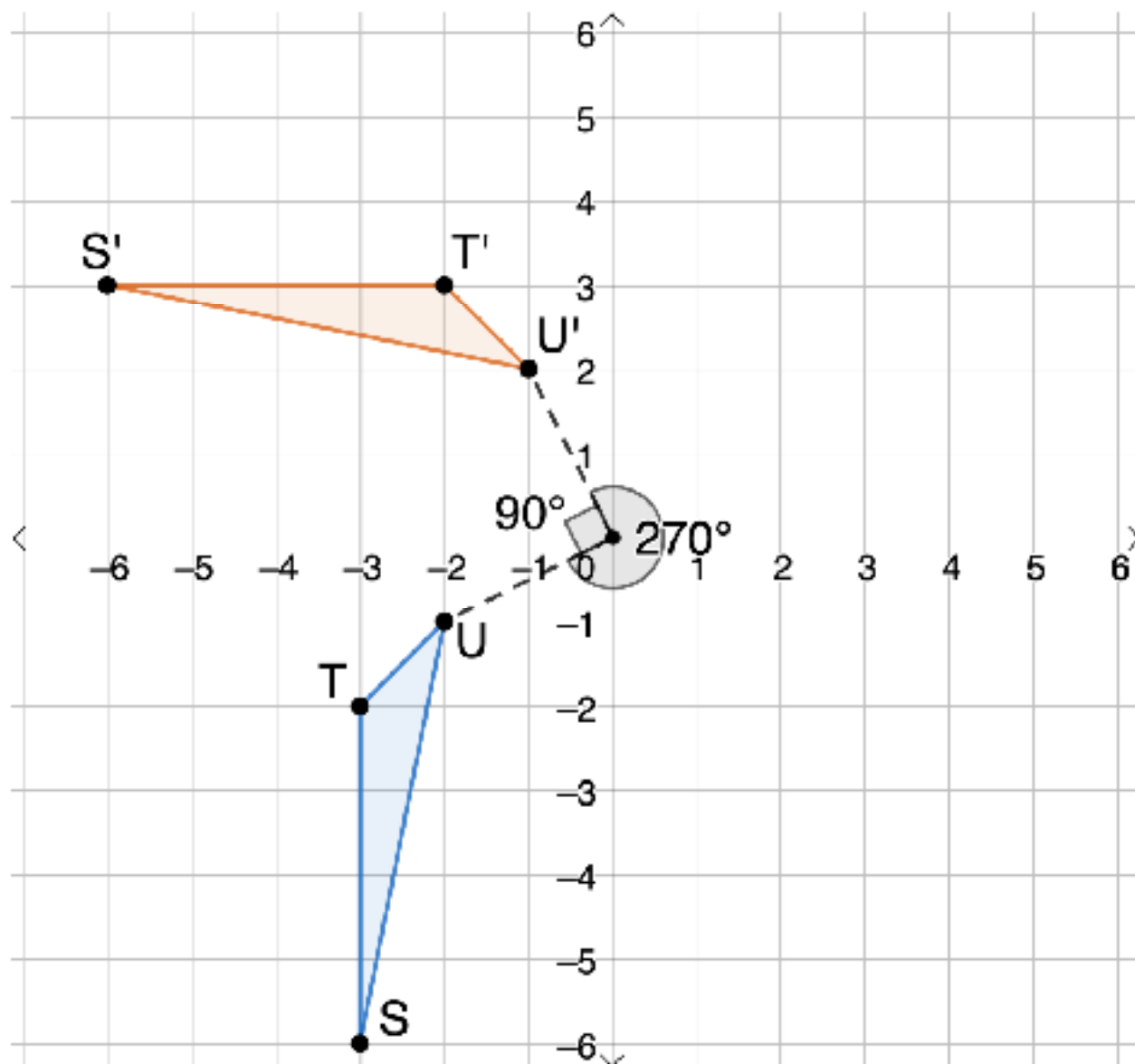
2. The diagram shows $\triangle GFL$ and $\triangle G'F'L'$ after a rotation about point R . Use a straightedge to draw $\angle GRG'$, $\angle FRF'$, and $\angle LRL'$.



3. Work with your partner to complete each statement using the angles formed by the corresponding vertices of each preimage and image and the center of rotation.
- Triangle $R'X'Z'$ is the result of rotating $\triangle RYZ$ _____° counterclockwise about point M .
 - Triangle GFL was rotated _____° counterclockwise about point R to create $\triangle G'F'L'$.

9.5.3 Guided Instruction: Rotations on the Coordinate Plane

1. On the coordinate grid below, $\triangle STU$ is rotated to create $\triangle S'T'U'$.



- a. There are two ways to describe the rotation. Complete both statements using the information given in the figure.

$\triangle STU$ was rotated _____ $^\circ$ clockwise about the origin to create $\triangle S'T'U'$.

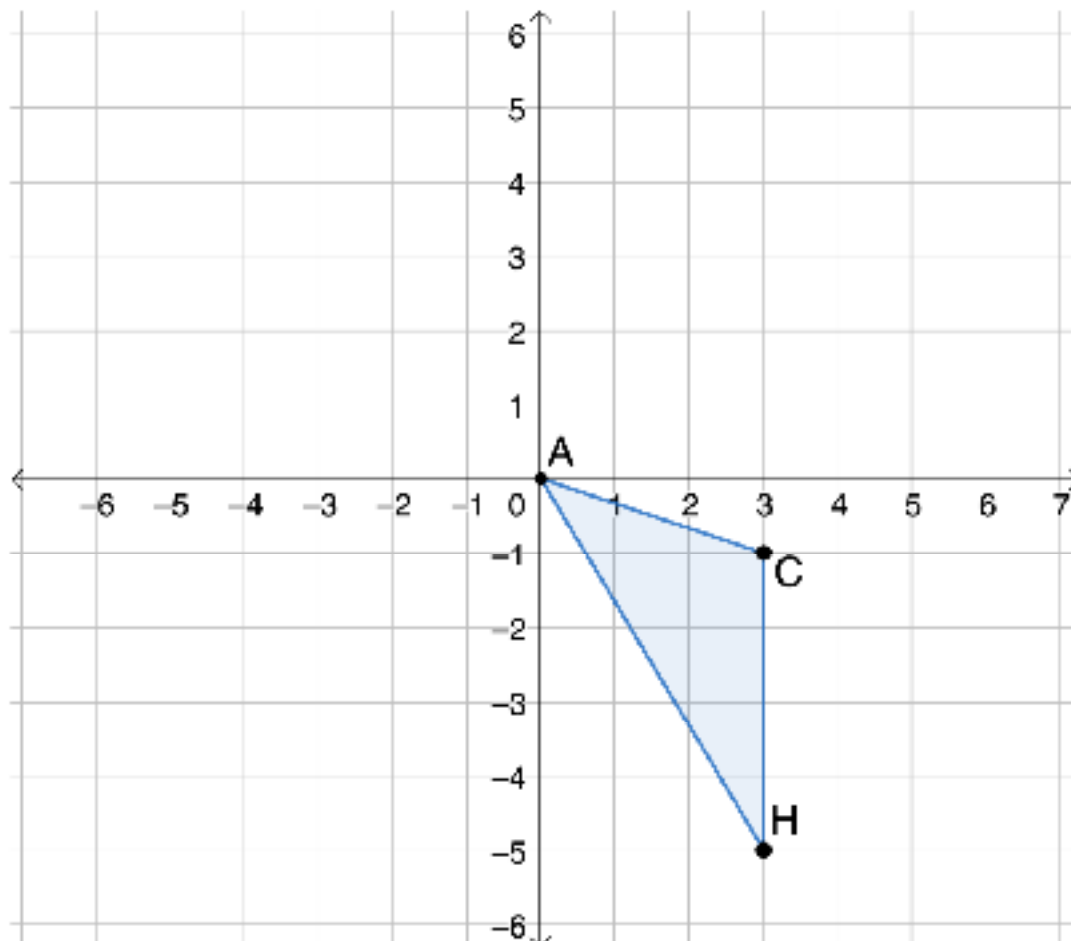
$\triangle STU$ was rotated _____ $^\circ$ counterclockwise about the origin to create $\triangle S'T'U'$.

- b. For every rotation clockwise, there is a rotation counterclockwise that would create the same image. Complete the table.

Clockwise Rotation	Equivalent Counterclockwise Rotation	Sum of Clockwise & Counterclockwise
90°		
180°		
270°		

- c. Explain what would happen if a figure were rotated 360° with the origin as the center of rotation.

2. Triangle ACH is shown on the coordinate grid.



Triangle $A'C'H'$ is the image created by rotating $\triangle ACH$ 180° clockwise about the origin.

- a. To show the rotation, follow these steps.
- Because a 180° angle is a straight line, use a straightedge to draw a dotted line from C through the origin that extends to the end of the grid.
 - Plot and label point C' on the dotted line so that $\overline{A'C'}$ is congruent to $\overline{AC}$.
 - Use a straightedge to draw a dotted line from H through the origin that extends to the end of the grid.
 - Plot and label point H' on the dotted line so that $\overline{H'A'}$ is congruent to $\overline{HA}$.
 - Connect the vertices to create $\triangle A'C'H'$.

- b. Describe how point A and point A' relate to each other.

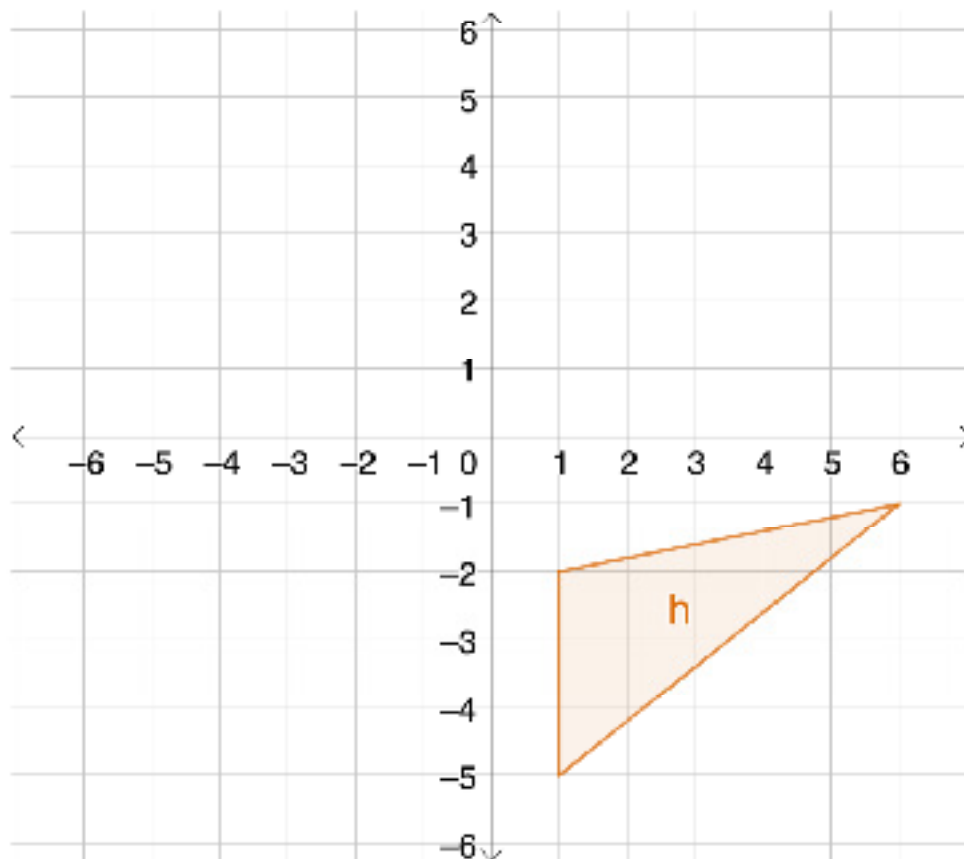
- c. Complete the table with the coordinates of each point.

	Coordinates		Coordinates
Point A		Point A'	
Point C		Point C'	
Point H		Point H'	

- d. What do you notice about the coordinates of the corresponding points in the preimage and the image after this rotation?

9.5.4 Collaboration: Which Quadrant?

1. A triangle is shown on the coordinate plane.



- a. Without performing the transformation, discuss with your partner what the image of the triangle would look like after a rotation 90° clockwise about the origin.

- b. Complete the statement.

The preimage of the triangle is in quadrant

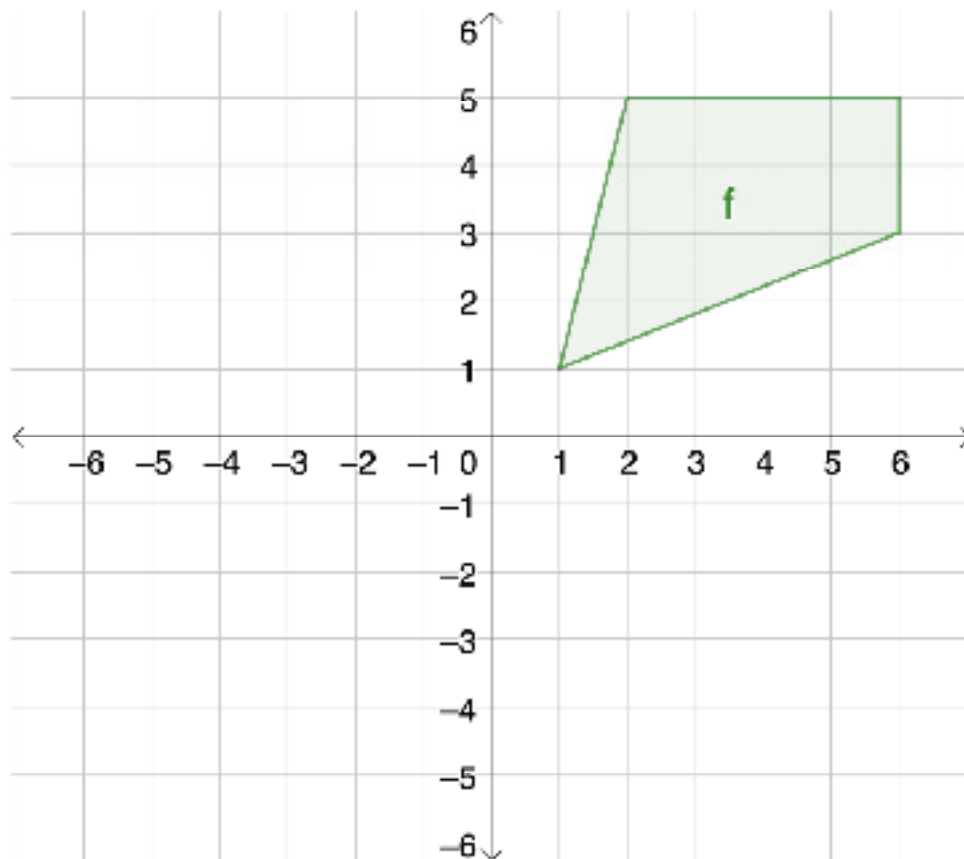
- ☐ 1.
- ☐ 2.
- ☐ 3.
- ☐ 4.

The image of the triangle after a 90° clockwise

rotation about the origin would lie in quadrant

- ☐ 1.
- ☐ 2.
- ☐ 3.
- ☐ 4.

2. A quadrilateral is shown on the coordinate plane.



Without performing the transformation, complete the statement.

The preimage of the quadrilateral is in quadrant

- ☐ 1.
- ☐ 2.
- ☐ 3.
- ☐ 4.

The image of the quadrilateral after a 270°

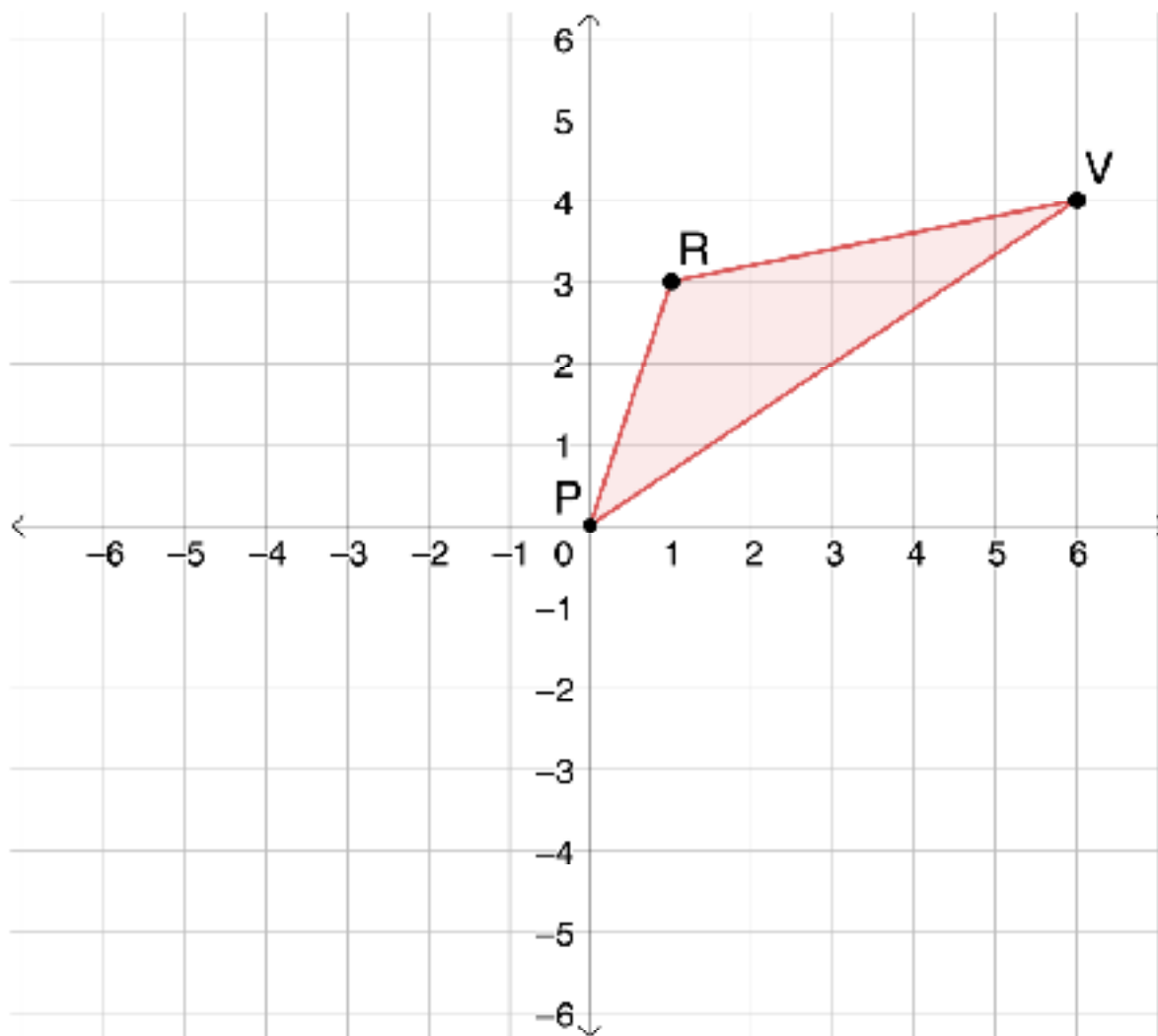
counterclockwise rotation about the origin would lie in

quadrant

- ☐ 1.
- ☐ 2.
- ☐ 3.
- ☐ 4.

9.5.5 Guided Practice: Rotations on the Coordinate Plane

1. Triangle PQR is plotted on the coordinate grid.



- a. Rotate $\triangle PQR$ 270° counterclockwise with the origin as the center of rotation. Label the image $\triangle P'Q'R'$.
- b. What is another way to describe the rotation performed in Part A?

9.5.6 Wrap-Up: Reflection

1. Take a **SELFIE** by checking the box that best matches what you did today.

Showed my work	☐ 	☐ 
Explained my answers	☐ 	☐ 
Lots of math vocabulary used	☐ 	☐ 
Found multiple solutions	☐ 	☐ 
I persevered through the problem	☐ 	☐ 
Eliminated careless errors	☐ 	☐ 

